

Xin Yao

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EDUCATION

Tianjin University of Science and Technology(TUST), Tianjin, CHN

09/2015-06/2019

- Major in Electronic Information Engineering
- Bachelor of Engineering
- Rank: 19/119
- Major GPA: 3.23

George Washington University, Washington DC, US

01/2020-06/2021

- Major in Electronic Engineering
- Focus on Communication and Networks
- Master of Science
- Major GPA: 3.4

University of Utah, Salt Lake City, Utah, US

08/2021-07/2024

- Major in Electronic Engineering
- Focus on Machine learning and the real-world wireless communication system
- Doctor of Philosophy
- Current GPA: 3.974

University of Florida, Gainesville, Florida, US

08/2024-now

- Major in Electronic Engineering
- Focus on Machine learning and the real-world wireless communication system
- Doctor of Philosophy
- Current GPA: 3.44

RESEARCH EXPERIENCE

Signal Processing and Software Design

11/2018-12/2018

Adviser: Professor Yang An (TUST)

Objective: Design automobile license plate recognition system based on MATLAB environment

Responsibilities:

- Design a special algorithm to capture the image of the license plate by surveillance cameras
- Using the filtering methods like hamming window to make the picture clear
- Design an algorithm to get the numbers and characters from the picture and matching them with the original data
- Show the match result on the screen and check it with the plate we have

Electronic Information System Comprehensive Experiment

11/2018-12/2018

Adviser: Professor Ya-Jun Li (TUST)

Objective: Design a music player based on Android system

Responsibilities:

- Built a basic environment that includes Android SDK, Eclipse, and JDK software to run.
- Used Eclipse software, Java language as the basic language, combined with machine assembly language

designed an Android music player.

- Compile XML and Java files separately and tested the application in practice with JDK simulation software;
- Complete the production of Android music players that support multiple file formats.

Curriculum Design of Electronic Technology

11/2017-01/2018

Adviser: Professor Yu-liang Liu (TUST)

Objective: Design a counter

Responsibilities:

- Designed the circuit diagram of counter through calculating amounts of theoretical knowledge.
- Searched for related suitable components, wire, and blank circuit receiving plate.
- Finished circuit connecting according to the circuit diagram.
- Tested the circuit by wiring up, and repaired fault circuits.

Design the voice control intelligent home system

01/2019-06/2019

Adviser: Lecturer Jing He (TUST)

Objective: Design the voice control home system based on STM32

Responsibilities:

- Design the hardware structures based on the STM32F103
- Connect the STM32F103 with the other instruments like speaker, PC and LCD screen
- Design an algorithm for LD3320 to collect the voice signal from the outside circumstance
- Assign all ports based on their abilities and write the codes to build the connection between STM32F103 and other parts of system like LD3320
- Run the code and test it by using the command like “open the light” or “run the light”
- Watch the feedback of the system like “the light is open” and adjust the code if the feedback is not right

“High-band Frequency Spectrum for 5G ” Presentation

03/2020-04/2020

Adviser: Professional Lecturer Sean Yun (GWU)

Objective: Introduce the 5G technology with the teamwork

Responsibilities:

- Introduce the basic 5G communication networks information
- Introduce the specific details about High-band Frequency Spectrum of 5G
- The protocols and the mathematics contents for High-band Frequency Spectrum of 5G
- Gathering all members’ work (other aspects of 5G) together and formed a final PPT

Encryption and Decryption for FTP

09/2020-10/2020

Adviser: Professor Tian Lan (GWU)

Objective: Encrypt and decrypt on the data packets under FTP by using Linux system.

Responsibilities:

- Design a special dialect to encrypt and decrypt the data
- Encrypt the data on the client side
- Decrypt the data (command) that received from client side on the server side
- Give the feedback to the client and finish the transmission
- Using the Wire-shark to confirm that the FTP transmission happened

Age of Information and Energy Harvest

09/2020-06/2021

Adviser: Professor Omur Ozel (GWU)

Objective: Focus on the AoI and the energy harvest process during the communication.

Responsibilities:

- Learn the details about the M/G/1/1 model and the idealize model with infinity buffer
- Learn the minimization methods of AoI from the previous research papers
- Learn the details about the M/G/1/2 with the packets deadlines and idealize model with infinity buffer
- Try to find the way that minimize the AoI in the M/G/1/1 model with deadlines and no buffer

Information Theoretic Secure Aggregation with Groupwise Keys

08/2021-06/2022

Adviser: Dr. Mingyue Ji (University of Utah)

Objective: Applications of different information theoretic secure aggregation algorithms.

Responsibilities:

- Learn the information theory on secure aggregation process
- Learn the python codes on parallel computing
- Apply different information theoretic algorithms

Federated Learning and Wireless Communication

08/2021-Now

Adviser: Dr. Mingyue Ji (University of Utah)

Objective: Focus on security on federated learning and the combination of machine learning and wireless communication.

Responsibilities:

- Design the new decentralized federated learning algorithm with error compensation.
- Simulations on Secure Aggregation algorithms.
- Implementation of Machine Learning algorithms.
- Combine the wireless communication system and ML algorithms.

Reinforcement Learning and Real-world Communication System

08/2023-Now

Adviser: Dr. Mingyue Ji (University of Florida)

Objective: Focus on reinforcement learning algorithm design and application, specifically in real-world testbeds

Responsibilities:

- Design the new state representation of reinforcement learning algorithm.
- Simulations on Dynamic control algorithms with different reinforcement learning algorithms.
- Deploy the reinforcement learning algorithms on real-world testbed

Develop and deploy the real-world Communication Testbed

08/2024-Now

Adviser: Dr. Mingyue Ji (University of Florida)

Objective: Focus on deployment of real-world testbed and corresponding algorithm testing

Responsibilities:

- Coding with C++ on modulation, demodulation, filtering, synchronization, encoding and decoding
- Python API connect to C++ application
- Automatic system design based on different physical layer devices

AWS Development

01/2026-Now

Adviser: Dr. Mingyue Ji (University of Florida)

Objective: Focus on AWS Web UI deployment

Responsibilities:

- Basic concepts of AWS and Certification Exam

- Coding skill about JSON, Java and JavaScript
- Web UI design based on AWS

Drone Assemble and Deployment with Wireless devices.

01/2026-Now

Adviser: Dr. Mingyue Ji (University of Florida)

Objective: Focus on drone assemble, flight controller adjustment and combine the communication device

Responsibilities:

- Physical connections of drone parts
- Flight controller initialization and parameter adjustment
- Drones assemble and customize payload design

INTERNSHIP EXPERIENCE

Tianjin Vocational Skills Training Base

Hexi, Tianjin, China

11/2018-12/2018

Position: Intern

- Handmade: Completed the connection of the crystal head of the network cable and the operation of three different methods of copper wire, and effectively controlled the intelligent building related to control system.
- Manual welding: Independently completed the flow lamp circuit board welding work of transistor, diodes, light-emitting diodes, resistance capacitors.
- 3D print design: Used NX drawing software for two-dimensional to three-dimensional(stretching, rotation, array mirroring and other three-dimensional operation) of the drawing task and output STL files.

Kings Island

Mason, OH, America

06/2018-09/2018

Position: Rides Associates

- Operated the recreational facilities with different electronic buttons and joysticks on the console.
- Assisted the maintainers in supervising and maintaining facilities in urgent circumstances.
- Tested these facilities under unmanned condition and manned condition respectively with normal operational process.

Anhui Sun Create Electronics Co. Ltd

Hefei, Anhui, China

07/2016-08/2016

Position: Intern

- Learned about how to supply C language into underlying database.
- Got a general idea about the future development of electronic specialty.

EXTRACURRICULAR ACTIVITIES

Person in charge of Sound and Lighting Department, TUST

2016-2018

- *Provided sound and lighting equipment for all kinds of events.*
- *Coordinated times and places with other departments.*

Performer and assistant in Graduation Party, TUST

2015-2018

- *Opened the show, along with self-organized MAC Band, with a medley of songs.*
- *Assisted in order keeping of the event.*
- *Assisted in seeking for sponsorship and organizing 2016 and 2017 Graduation Party.*

Member of Symphony Orchestra, Art Troupe, TUST

2015--2016

- *Took part in all kinds of trainings, rehearsals, and shows.*

SKILLS

MS Excel/PowerPoint/Word/Publisher; Adobe Premiere Pro/Adobe After Effects; Adobe Acrobat /Photoshop; C, XHTML, DHTML; NX 8.0; Eclipse, Android SDK(JDK); MATLAB; Auto CAD; Python; C++; JSON; JavaScript;

Java; AWS; Sharp3D

HONORS & AWARDS

Excellent Leader of Sound and Lighting department	2018
Third-prize Scholarship of TUST	2017
Second Prize of personal project (percussion music) in Art Performance, TUST	2016
First Prize of group project (orchestra music) in Art Performance, TUST	2016
Advanced Individual of Social Work in TUST	2016
Excellent League Member of TUST	2016